

POWERACT

CONSULTING

INSTALLATION GUIDE

DynamicMS — Installation Guide

Server and web client, demonstration data in English, French and Arabic

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What you install

POWERACT Consulting delivers DynamicMS as one zip file with two folders. Installation needs only Node.js: the database is the SQLite engine built into Node.js, so no database server, compiler or container is required.

Folder	Content	Port
server	REST API (Node.js, Express), SQLite database, seed generator, reports (PDF, Excel, Word, CSV), backups, licence tools, tests.	4000
web	Web client (React, Vite). In development it proxies /api to the server; built, it is served by the server.	5173

Requirements

Item	Minimum
Operating system	Windows 10/11, macOS 13 or later, or a recent Linux distribution
Node.js	Version 22.13 or later (LTS 22 or 24). The built-in node:sqlite module is required.
npm	Installed with Node.js
Disk space	1.5 GB (dependencies about 400 MB, demonstration database about 190 MB, backups)
Memory	2 GB free
Browser	Current Chrome, Edge, Firefox or Safari (desktop, tablet or mobile)
Network	Internet access for npm install only; the application itself runs offline

Install Node.js

- **Windows and macOS.** Download the LTS installer from nodejs.org, run it with the default options, then open a new terminal.
- **Linux.** Use the NodeSource packages or nvm, for example: `nvm install 22`.
- **Check.** Type `node --version`. The result must be v22.13.0 or later.

Install and start the application

1. Unzip

Unzip DynamicMS.zip into a folder of your choice, for example C:\DynamicMS or ~/DynamicMS. You obtain the folders server and web.

2. Server: install, seed, start

Open a terminal in the server folder and type:

Command

```
cd server
```

Command

npm install

npm run seed

npm run dev

- **npm install** downloads the server dependencies (about one minute).
- **npm run seed** creates data/dynamicms.db with the full demonstration: 2 groups and 1 independent organization, 60 organizations, 120 full-run projects (QMS and QHSE for each vertical, large companies and SMEs), 1,141 users and every record of the runs, in English, French and Arabic. It takes 10 to 30 seconds and can be run again at any time to reset the data.
- **npm run dev** starts the API on <http://localhost:4000> and restarts it when the code changes. Check <http://localhost:4000/api/health>: it returns status ok.

3. Web client: install and start

Open a second terminal in the web folder and type:

Command

cd web

npm install

npm run dev

Open <http://localhost:5173> in your browser. Keep both terminals open while you use the application.

4. Sign in

Account	E-mail	Password
Platform administrator	admin@dynamicms.example	Admin@2026
Universal large company — Quality Manager	quality@horizon-universal.example	Demo@2026
Universal large company — IMS Manager	ims@horizon-universal.example	Demo@2026
Universal SME — IMS Manager	ims@atlas-sme.example	Demo@2026
AEC & Construction SME — Quality Manager	quality@nova-aec.example	Demo@2026
Group top management (automotive)	ceo@horizon-aut.example	Demo@2026

Every organization has 19 accounts named `role@domain`: `ims`, `quality`, `hse`, `risk`, `compliance`, `audit`, `hr`, `documents`, `it`, `operations`, `performance`, `esg`, `transformation`, `process`, `admin`, `ceo`, `employee`, `aigov` and `auditor`. Large companies use `horizon-<vertical>.example` (for example `horizon-aec.example`), SMEs `nova-<vertical>.example`, the universal holding `horizon-universal.example` and the independent SME `atlas-sme.example`.

Configuration

The server reads the following environment variables. Defaults suit a local installation.

Variable	Default	Purpose
PORT	4000	API port
DB_FILE	server/data/dynamicms.db	SQLite database file
JWT_SECRET	development value	Signing key of sessions. Mandatory in production: the server refuses to start with the default.
JWT_TTL	12h	Session duration
CORS_ORIGIN	http://localhost:5173	Origins allowed to call the API (comma-separated)
BACKUP_DIR	server/backups	Backup folder
BACKUP_RETENTION_DAYS	14	Backups older than this are deleted
STORAGE_DIR	server/storage	Attached files
DEPLOYMENT_MODE	saas	saas or onprem (on-premises requires a signed licence)
LICENCE_FILE	server/licence/licence.lic	Licence file for on-premises mode

Production deployment

1. Build the web client: in the web folder type `npm run build` (creates `web/dist`).
2. In the server folder, set `JWT_SECRET` to a long random value and `NODE_ENV=production`.
3. Start the server with `npm start`. It serves the API and the built web client on the same port (`http://<host>:4000`).
4. Place a reverse proxy (IIS, nginx, Apache) with TLS in front of port 4000 for access over the network.

Command

```
cd web && npm run build
cd ../server
set JWT_SECRET=<long random value> (Windows) | export JWT_SECRET=<long random value> (macOS, Linux)
set NODE_ENV=production | export NODE_ENV=production
npm start
```

Backups and restore

The server takes a consistent backup when it starts (if the last one is older than a day) and then every 24 hours, and deletes backups older than the retention period. You can also back up on demand:

Command

```
npm run backup
npm run backup -- --list
```

Command

```
npm run backup -- --verify dynamicms-<date>-cli.db
```

To restore: stop the server, copy the chosen backup file over server/data/dynamicms.db, then start the server again. The platform administrator can also back up from Administration › Operations.

On-premises licence

In on-premises mode the server checks a licence file signed with an Ed25519 key: customer, number of seats, expiry date and packs. Without a valid licence the platform stays readable but refuses changes.

Command

```
npm run sign-licence -- --customer "Horizon Industrial Group" --seats 1500 --expires 2027-12-31 --packs DMS-ENT,DMS-AI
```

```
set DEPLOYMENT_MODE=onprem (Windows) | export DEPLOYMENT_MODE=onprem
```

```
npm start
```

Security. The first signing creates the key pair in server/licence. Keep private.pem with the vendor; the customer server needs only public.pem and licence.lic.

AI models (optional)

The AI use cases work without any external service: the built-in engine answers from the DynamicMS content and the organization data. To use a large language model instead, the tenant administrator opens Administration › AI models, chooses a standard provider and model (Anthropic Claude, OpenAI, Azure OpenAI, Google Gemini, Mistral) or a custom model with an OpenAI-compatible HTTPS endpoint, types the API key, selects Test the connection and enables it.

- **Keys.** API keys are encrypted at rest (AES-256-GCM, key derived from JWT_SECRET) and never sent back to the browser. Changing JWT_SECRET makes stored keys unreadable: type them again.
- **Network.** The server calls the provider from its own network: allow outbound HTTPS to the provider host (for example api.anthropic.com) in the firewall or proxy.
- **Fallback.** If the provider does not answer within 30 seconds or returns an error, the built-in engine answers and the user sees a warning.

Document layout and templates

Generated documents (Word, PDF, Excel) use the layout of the organization: Records › Documents › Layout (logo image in PNG or JPEG up to 2 MB, colors, header and footer). Logos are stored in

STORAGE_DIR. The 37 IMS templates are delivered with the application; organizations copy and adapt them in Records › Documents › Templates.

Automated tests

In the server folder, npm test runs the 12 API tests on a copy of the seeded database: sign-in, full-run step completion and reopen, gate rules, cross-tenant isolation, RBAC, one-Accountable RACSI rule, owner-evaluator separation, report exports in three languages and four formats, AI suggestions, benchmarking, project creation modes, permission matrix, compliance disclosure, backups, and the structured forms (record tables, actions from plans, SMART objectives, KPI from a step), document templates (generation, structure, versions, downloads, layout), RACSI per macro process and step, readiness checklist, AI model settings and group and organization creation.

Command

```
cd server
```

```
npm test
```

Troubleshooting

Message or symptom	Cause and remedy
“The database is empty. Run npm run seed first”	Run npm run seed in the server folder, then npm run dev.
“Cannot find module node:sqlite” or “No such built-in module”	Node.js is older than 22.13. Install the current LTS and open a new terminal.
“EADDRINUSE: port 4000 (or 5173) already in use”	Another program uses the port. Stop it, or start the server with PORT=4100 and update the proxy in web/vite.config.js.
The web page shows “Something went wrong” after sign-in	The server is not running. Start it with npm run dev in the server folder.
npm install fails behind a company proxy	Configure npm: npm config set proxy http://proxy:port and npm config set https-proxy http://proxy:port.
You want to start again with clean demonstration data	Stop the server, run npm run seed, start the server.